MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS PURE MOLYBDENUM, TYPE I, CLASS 2, COMPOSITION A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Mo	O_2	Others (1)	
		Each	Total
99.5 min.	0.15 max.	-	0.5 max.

(1) Determination not required for routine acceptance.

Particle Size/Distribution

+170 Mesh (+90 µm)	1.0 max.
+200 Mesh (+75 µm)	15.0 max.
+325 Mesh (+45 µm)	80.0 min.

Sprayed Coating Properties

Bond Strength	41.4 MPa (6000 psi) min.
Hardness	Rockwell 15N, 75 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

AMSC N/A FSC 3439

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Deposit Efficiency and Deposition Rate

70% min for N_2/H_2 @ 9.5 lb/hr
50% min for A/H_2 @ 3.5 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF41, Class A
Pratt & Whitney Aircraft	PWA 1313
Rolls-Royce, Ltd.	MSRR 9507/19

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/09)

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